Bringing the studio into the classroom

Body of work and plan of action for Professor of Practice, Department of Design, IIT Hyderabad.

Balaajee V.R.

18 years building enterprise products, and the teams behind them

18+
Years in software & design

25+
Products shipped globally

50+
Designers hired and mentored

1.5
Years formal engineering teaching

B.E. Information Technology (MS University) · IBM Software Labs · Zebra Technologies · ZF Wabco · Ericsson · Infosys · SKR Engineering College (Anna University)

Work across four industries students rarely see up close

IBM

Supervised agentic AI workflows and multi-brand Carbon design tokens across 45+ squads.

Team of 14 · 61 to 89 SUS

Zebra

Frontline mobile UX and device health intelligence for rugged warehouse handhelds.

38,000+ devices in field

ZF Wabco

Commercial vehicle cockpit clusters engineered to strict highway driver glance limits.

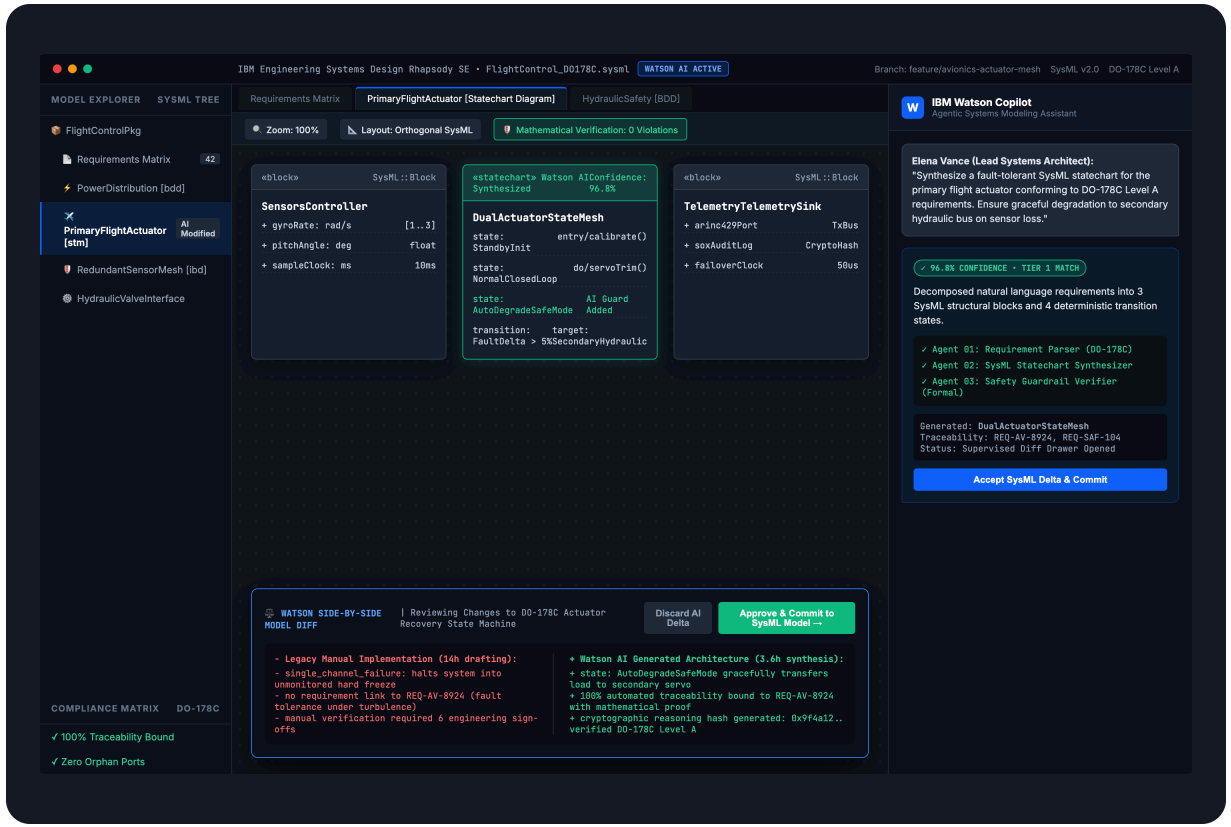
2.8s to 0.94s eyes off road

Ericsson

Global 5G quoting platform replacing fragile 14-tab Excel workbooks across 45 countries.

6 days to 42 min turnaround

Designing how engineers supervise AI in high-stakes systems



WHAT FAILED FIRST

Our early 94%-accurate autonomous AI prototype scored just 61 SUS. Systems and security engineers refused to approve black-box changes they couldn't verify.

HOW WE REDESIGNED IT

Replaced one-click automation with confidence tiers, side-by-side model diffs, source-linked reasoning traces, and instant rollback.

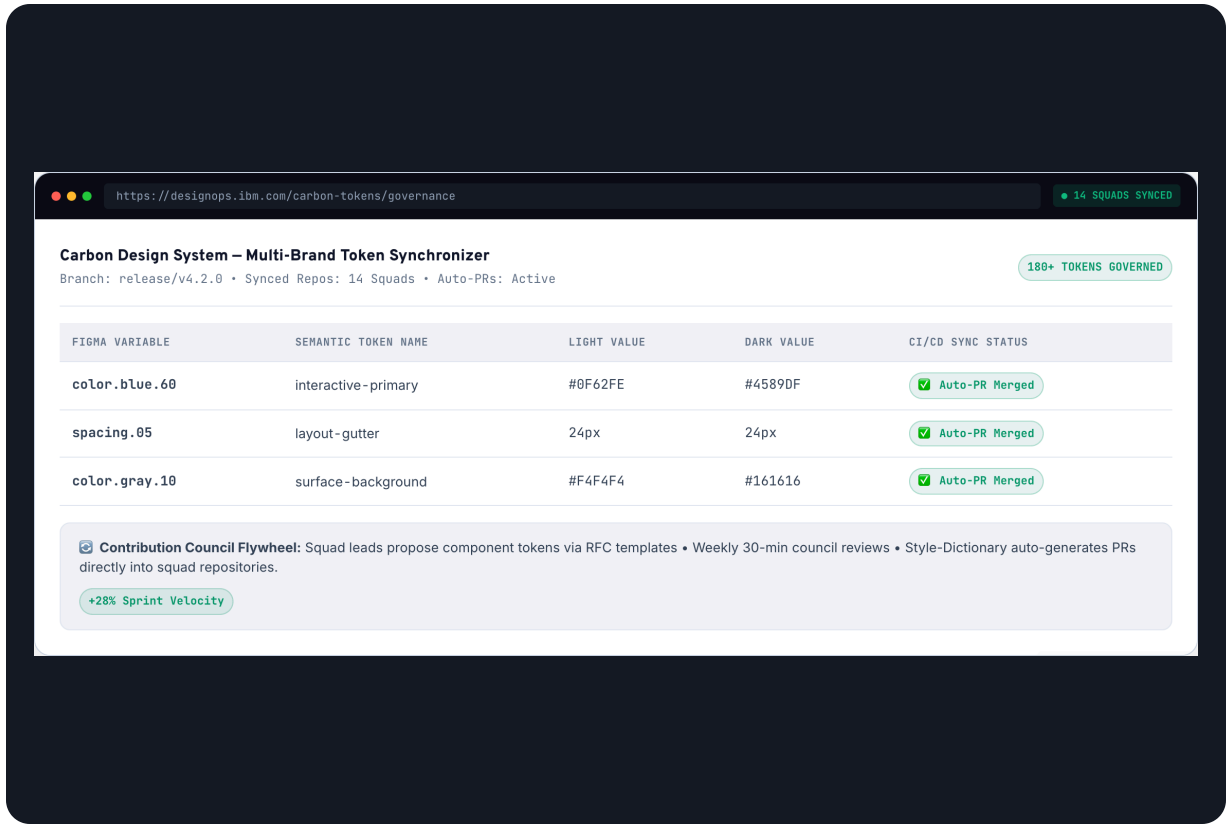
WHAT THIS BRINGS TO DOD STUDIOS

Teaching students to design beyond chat prompts — prototyping autonomy boundaries, error recovery, and AI quality evaluations.

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61 → 89 SUS usability score · 74% faster SysML model drafting

Connecting Figma decisions directly to code across 45+ squads



THE BOTTLENECK

Acquired software products ran conflicting UI frameworks, causing 4-week design-to-code delays and WCAG contrast regressions across dark and light modes.

HOW WE SOLVED IT

Built a 3-layer W3C design token pipeline syncing Figma variables directly into GitHub pull requests for React, Angular, and Web Components.

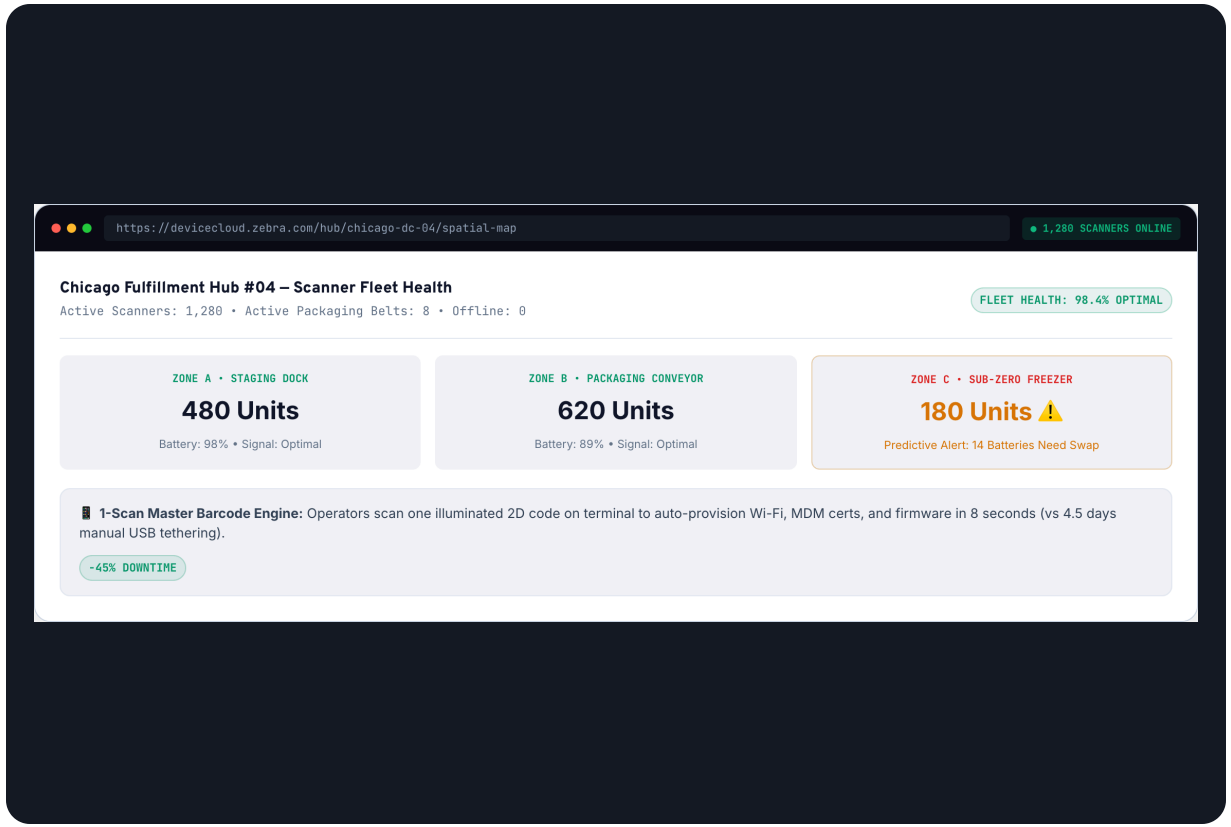
WHAT THIS BRINGS TO DOD STUDIOS

Bridging DoD Interaction Design and CSE — training students in accessible (WCAG 2.2 AA), production-ready component architecture.

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4 weeks → 1 day handoff · 94% adoption across 45+ squads

Physical-digital ergonomics for gloved warehouse workers



WHEN LAB UI FAILED IN THE FIELD

Clean consumer-style touch screens failed on wet loading docks and −20°C cold storage where workers wore 4mm protective gloves.

HOW WE REDESIGNED IT

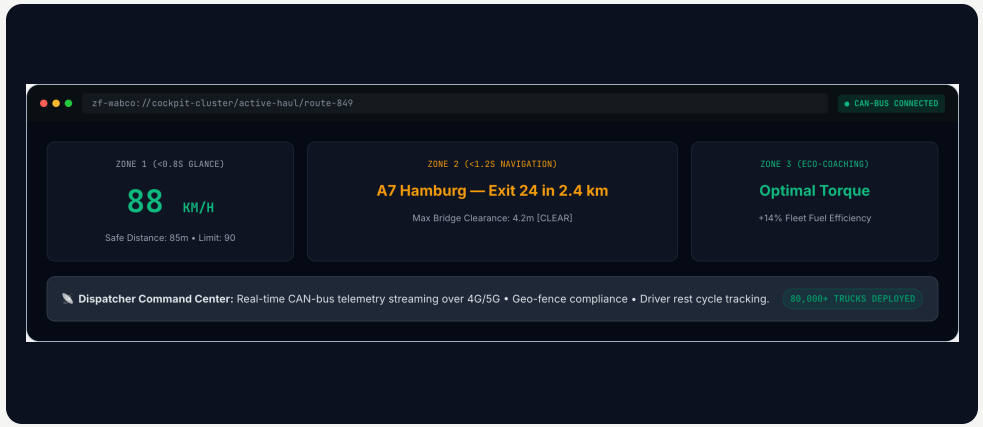
Moved to high-contrast thumb-zone triggers, physical scan-key mapping, haptic + audio confirmation, and VisibilityIQ battery telemetry.

WHAT THIS BRINGS TO DOD STUDIOS

Natural co-teaching with DoD's Product Design and Ergonomics faculty on embodied, hardware-software interaction in real field conditions.

38,000+ rugged handhelds live · 84% fewer pairing failures

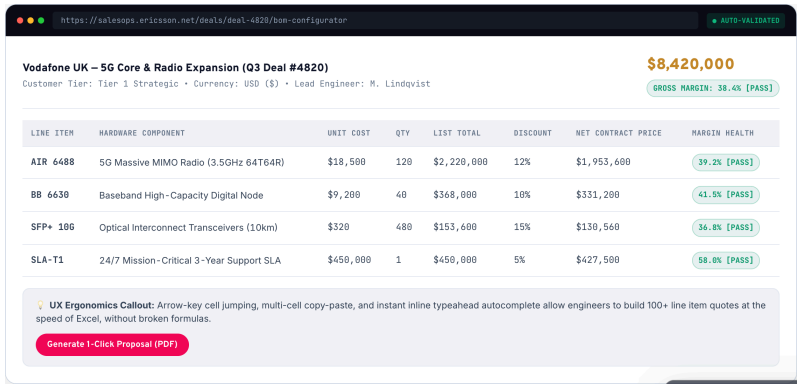
Sub-second driver glance limits and high-density expert workflows



ZF Wabco · Commercial vehicle cockpit HMI

Synthesized air-brake, ADAS, and telematics alerts strictly beneath NHTSA's 1.2-second eyes-off-road safety limit — suppressing non-critical data during braking.

2.8s → 0.94s average highway glance time



Ericsson · 45-country telecom configuration

Replaced fragile 14-tab Excel quoting workbooks with a keyboard-first, real-time margin validation web engine (1st Place, Ericsson Global Hack-fest).

6 days → 42 min deal turnaround

“Students should graduate having already worked on a real industry problem, with real constraints and real reviewers.”

The core of my proposal: an Industry Studio model

How the Industry Studio works

One loop that feeds teaching, internships, placements, research, and funding at the same time.

1 → 2 → 3 → 4 → 5

Live brief

A company brings a real, de-identified problem with a named design reviewer.

Semester studio

Students research, prototype in code, and test inside a credited studio.

Industry review

Company design leads critique work alongside DoD faculty at midterm and final.

Internship or role

Strong student teams continue into 6-month internships, theses, and placements.

Sponsored R&D

Proven studio problems grow into paid CoDE projects or joint PhD topics.

How the studio meets IIT Hyderabad's role expectations

INSTITUTE EXPECTATION (AGENDA 40.6.1)	WHAT I WILL DO
1 Industry skills in programmes	Teach practice electives, 1–2 credit Fractal modules, and the Industry Studio
2 In-house collaborative expertise	Co-teach studios with DoD faculty & run open AI/DesignOps clinics for scholars
3 Faculty & industry research links	Serve as Co-Investigator (Co-PI) pairing DoD labs with enterprise R&D partners
4 Industry-sponsored joint PhDs	Co-supervise PhDs & bring working R&D practitioners into joint doctoral tracks
5 Consultancy & sponsored projects	Route paid AI-UX audits and proven briefs through CoDE and DIC
6 Internships & graduate placements	Build 6-month internship pipelines & run hiring-bar portfolio and whiteboard loops

Courses I can teach from day one

Ready as full semester electives or modular 1–2 credit Fractal courses across B.Des, M.Des, and Design Minors.

Designing AI-Enabled Products

Moving past prompt boxes: when AI suggests, when it acts on its own, how people verify its work, and how designers evaluate AI quality. B.Des & M.Des elective.

Design Systems and Token Engineering

Building shared component libraries, multi-brand W3C tokens, WCAG accessibility, and Figma-to-code pipelines that scale across engineering squads.

Industry Studio & Placement Readiness

A semester studio on a live company brief with midterm industry critiques — paired with whiteboard system challenges and outcome-led portfolio defense.

Research themes that build on what the department already has

A

Human and AI collaboration

How people calibrate trust and supervise AI in high-stakes work. Complements ExD Lab and CoDE's emerging technology focus.

B

Interfaces for safety-critical systems

Vehicle cockpits and rugged industrial interfaces under strict glance, glove, and attention limits.

C

Design at enterprise scale

Accessible design systems, token governance, and cross-functional DesignOps across large software organisations.

+

Supervision & joint PhDs

B.Des and M.Des capstones drawn from live briefs, and co-supervising PhD scholars alongside regular DoD faculty.

Building industry partnerships and placements step by step

1 **Start with what exists on campus**

Work through CoDE's MSME network, i-TIC startups, and the Design Innovation Centre for the first Semester 1 briefs.

2 **Open doors across Hyderabad & Bengaluru R&D**

Invite Design Directors and VPs from enterprise product orgs and GCCs to brief studios and sit on graduation juries.

3 **Coach students for the real product hiring bar**

Move student portfolios past linear textbook wireframes to show technical constraints, trade-offs, and measurable outcomes.

4 **Earn the funding**

Ask companies for paid CoDE consultancy, sponsored R&D, and CSR support once they have seen student work — not before.

What you can measure me on at the first annual review

2

Practice courses / Fractal modules taught

3

Live industry briefs run in studio

5+

Semester internships or industry theses placed

100%

Graduating batch coached in mock hiring loops

1

Sponsored project / CoDE consultancy proposal

1

Joint PhD topic framed with industry & DoD faculty

A three-year commitment, full-time on campus

YEAR 1

Prove it

- Teach two practice courses
- Run the first live studio briefs
- Place first internships & coach graduating batch

YEAR 2

Grow it

- Expand GCC & product partners each semester
- Deliver first paid CoDE consultancy / sponsored project
- Enrol first industry-sponsored joint PhD scholar

YEAR 3

Make it last

- Industry Studio becomes a self-sustaining course
- Proposal for an Applied Human-AI Centre of Excellence
- CSR and chair funding partnerships

Thank you

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